Variance Reduction in Monte Carlo: Theoretical Foundations

Quant Finance Portfolio — Phase 2, Block 2.0

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Abstract

We develop the two foundational variance reduction techniques used throughout the rest of Phase 2: antithetic variates and control variates. For each we derive the conditions under which it is guaranteed to reduce variance, the resulting variance formula, and the practical algorithm that implements it. The treatment is honest about the cost-benefit trade-off: a method that halves the variance but doubles the per-path cost is no improvement, and we make the *variance times time* (VxT) criterion explicit at the outset to guard against this confusion. We also derive the empirical control-variate coefficient and analyse the small bias it introduces, and we close with a brief catalogue of techniques we do not cover in the present block (stratified sampling, importance sampling, multilevel MC) together with the reasons.

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1 Introduction

Every Monte Carlo estimator we have built so far has had the same asymptotic structure: $\hat{I}_M = M^{-1}\sum_{i=1}^{M} Y_i$ for some i.i.d. payoff Y, with variance $\mathrm{Var}(\hat{I}_M) = \sigma_Y^2/M$ where $\sigma_Y^2 = \mathrm{Var}(Y)$. The half-width of a fixed-confidence interval scales as $\sigma_Y/\sqrt{M}$.

For a typical at-the-money European call (parameters $S_0 = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$) the payoff variance is $\sigma_Y^2 \approx 217$. A half-width of 0.01 at 95% confidence requires

$M \approx 4 \cdot 1.96^2 \cdot 217/0.01^2 \approx 8.3 \cdot 10^6$ paths. Each ten-fold reduction in the half-width costs a hundred-fold increase in the sample size.

Variance reduction techniques attack this rate by constructing *alternative* estimators of the same quantity $I = \mathbb{E}[Y]$, with the same sample size M, but with variance smaller than σ_Y^2/M. The reduction comes from exploiting structure of the problem (monotonicity, correlation with a tractable quantity) that the basic estimator ignores.

The general framework is: replace Y with another random variable $\tilde{Y}$ such that $\mathbb{E}[\tilde{Y}] = \mathbb{E}[Y]$ but $\mathrm{Var}(\tilde{Y}) < \mathrm{Var}(Y)$, and run the Monte Carlo with $\tilde{Y}$ in place of Y. The estimator $\hat{I}_M^{\tilde{Y}} = M^{-1}\sum_i \tilde{Y}_i$ has variance $\mathrm{Var}(\tilde{Y})/M$. If $\mathrm{Var}(\tilde{Y}) = \sigma_Y^2/k$, the variance is reduced by a factor of k, which is equivalent to multiplying the effective sample size by k.

The catch, almost always, is that producing one sample of $\tilde{Y}$ costs more than producing one sample of Y. The honest way of comparing is the *variance times time* criterion of Section 2.

2 The variance-times-time (VxT) criterion

Suppose two estimators of the same quantity I are available:

- Estimator A: each sample has variance σ_A^2 and takes time τ_A to compute.
- Estimator B: each sample has variance σ_B^2 and takes time τ_B to compute.

Given a total computational budget T_{total}, estimator A can produce $M_A = T_{\text{total}}/\tau_A$ samples, leading to estimator variance $\sigma_A^2/M_A = \sigma_A^2\tau_A/T_{\text{total}}$. Similarly, B produces estimator variance $\sigma_B^2\tau_B/T_{\text{total}}$.

The estimator with smaller *product* $\sigma^2\tau$ wins, and the relative VxT,

$$\frac{\sigma_A^2\tau_A}{\sigma_B^2\tau_B}, \tag{1}$$

quantifies how many times faster B converges to the truth at fixed budget. A variance reduction technique that halves the variance but doubles the per-sample cost is a wash: VxT unchanged, no actual improvement.

This criterion will reappear in every validation script of Phase 2 Block 2. Variance ratios alone are misleading; only the VxT ratio captures the practical comparison.

Remark 2.1. The VxT criterion assumes the per-sample cost is the same for every sample, which is true for the simple estimators considered here. For adaptive estimators (where the cost varies with the realisation), a more careful analysis is needed; see Glasserman [3], Section 4.1.3.

3 Antithetic variates

3.1 Setup and theorem

Suppose Y depends on a vector of standard normals $\mathbf{Z}$: $Y = f(\mathbf{Z})$ with $\mathbf{Z} \sim \mathcal{N}(0, I)$. By symmetry of the normal distribution, $-\mathbf{Z}$ is also $\mathcal{N}(0, I)$, so $f(-\mathbf{Z})$ has the same distribution (and the same expectation) as $f(\mathbf{Z})$.

The antithetic variates estimator pairs the two:

$$Y_i^{\mathrm{AV}} \stackrel{\text{def}}{=} \tfrac{1}{2}\big(f(\mathbf{Z}_i) + f(-\mathbf{Z}_i)\big), \qquad i = 1, \ldots, M, \tag{2}$$

with $\mathbf{Z}_i$ i.i.d. standard normals across i. The estimator of $I = \mathbb{E}[f(\mathbf{Z})]$ is

$$\hat{I}_M^{\mathrm{AV}} \stackrel{\text{def}}{=} \frac{1}{M}\sum_{i=1}^{M} Y_i^{\mathrm{AV}}. \tag{3}$$

Theorem 3.1 (Variance of the antithetic estimator)**.** *The estimator $\hat{I}_M^{\mathrm{AV}}$ is unbiased for I, and its variance equals*

$$\mathrm{Var}\big(\hat{I}_M^{\mathrm{AV}}\big) \;=\; \frac{1}{2M}\big(\,\mathrm{Var}(f(\mathbf{Z})) + \mathrm{Cov}(f(\mathbf{Z}),\, f(-\mathbf{Z}))\,\big). \tag{4}$$

In particular, $\mathrm{Var}(\hat{I}_M^{\mathrm{AV}}) < \mathrm{Var}(\hat{I}_M^{\mathrm{IID}})$ *if and only if* $\mathrm{Cov}(f(\mathbf{Z}), f(-\mathbf{Z})) < \mathrm{Var}(f(\mathbf{Z}))$*, equivalently* $\mathrm{Corr}(f(\mathbf{Z}), f(-\mathbf{Z})) < 1$*, which is the generic case.*

Proof. Unbiasedness is immediate: $\mathbb{E}[Y_i^{\mathrm{AV}}] = \frac{1}{2}(\mathbb{E}[f(\mathbf{Z})] + \mathbb{E}[f(-\mathbf{Z})]) = \mathbb{E}[f(\mathbf{Z})] = I$. For the variance, $\mathrm{Var}(Y_i^{\mathrm{AV}}) = \frac{1}{4}(\mathrm{Var}(f(\mathbf{Z})) + \mathrm{Var}(f(-\mathbf{Z})) + 2\,\mathrm{Cov}(f(\mathbf{Z}), f(-\mathbf{Z})))$, and $\mathrm{Var}(f(-\mathbf{Z})) = \mathrm{Var}(f(\mathbf{Z}))$ by distributional equality, so $\mathrm{Var}(Y_i^{\mathrm{AV}}) = \frac{1}{2}(\mathrm{Var}(f(\mathbf{Z})) + \mathrm{Cov}(f(\mathbf{Z}), f(-\mathbf{Z})))$. The estimator's variance is this divided by M. □

The strict comparison with the IID estimator is more informative when made at *equal computational cost*, since each Y_i^{AV} requires two evaluations of f rather than one. The honest comparison is between $\hat{I}_M^{\mathrm{AV}}$ and $\hat{I}_{2M}^{\mathrm{IID}}$, both using $2M$ payoff evaluations:

$$\mathrm{Var}\big(\hat{I}_{2M}^{\mathrm{IID}}\big) \;=\; \frac{\mathrm{Var}(f(\mathbf{Z}))}{2M}. \tag{5}$$

The AV estimator wins if and only if $\mathrm{Cov}(f(\mathbf{Z}), f(-\mathbf{Z})) < 0$. This brings us to the key sufficient condition.

3.2 The monotonicity condition

Theorem 3.2 (Sufficient condition for AV to reduce variance)**.** *If $f : \mathbb{R}^d \to \mathbb{R}$ is monotone in each coordinate of $\mathbf{Z}$ (in the same direction, i.e. either monotone increasing in all or monotone decreasing in all), then*

$$\mathrm{Cov}(f(\mathbf{Z}),\, f(-\mathbf{Z})) \le 0, \tag{6}$$

with strict inequality unless f is essentially constant.

This is a special case of a general result on monotone functions of random vectors and antithetic transformations; see Glasserman [3, Section 4.2]. We do not reproduce the proof; the intuition is that if f is monotone, large values of $f(\mathbf{Z})$ correspond to favourable directions of $\mathbf{Z}$, and reflecting $\mathbf{Z}$ to $-\mathbf{Z}$ flips these to unfavourable directions, producing small values of $f(-\mathbf{Z})$. The pair $(f(\mathbf{Z}), f(-\mathbf{Z}))$ is therefore negatively correlated.

For the European call under GBM, $S_T = S_0 \exp((r - \sigma^2/2)T + \sigma\sqrt{T}Z)$ is strictly increasing in Z, and the call payoff $(S_T - K)^+$ is monotone increasing in S_T, hence in Z. So the monotonicity condition holds and AV is guaranteed to reduce variance for the European call. By how much depends on how strong the negative correlation is, which we measure empirically in Block 2.1.

4 Control variates

4.1 Setup and theorem

Suppose, in addition to the target Y, we have access to a second random variable X defined on the same paths, with the property that $\mathbb{E}[X]$ is *known in closed form.* The variable X is called a *control variate.*

For any constant $c \in \mathbb{R}$, the centred quantity $X - \mathbb{E}[X]$ has mean zero, so the random variable

$$\tilde{Y}(c) \;\stackrel{\text{def}}{=}\; Y - c\,(X - \mathbb{E}[X]) \tag{7}$$

satisfies $\mathbb{E}[\tilde{Y}(c)] = \mathbb{E}[Y] - c \cdot 0 = \mathbb{E}[Y] = I$. The estimator built from $\tilde{Y}(c)$ is unbiased for I regardless of c. The freedom to choose c lets us reduce variance.

Theorem 4.1 (Optimal control coefficient)**.** *The variance of $\tilde{Y}(c)$ is*

$$\mathrm{Var}(\tilde{Y}(c)) \;=\; \mathrm{Var}(Y) - 2c\,\mathrm{Cov}(Y,X) + c^2\,\mathrm{Var}(X), \tag{8}$$

minimised at

$$c^* \;\stackrel{\text{def}}{=}\; \frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}, \tag{9}$$

with minimum value

$$\mathrm{Var}(\tilde{Y}(c^*)) \;=\; \mathrm{Var}(Y)\,(1-\rho^2), \qquad \rho \;\stackrel{\text{def}}{=}\; \mathrm{Corr}(Y,X). \tag{10}$$

Proof. The variance formula is a straightforward expansion of $\tilde{Y}(c)$. Setting the derivative with respect to c to zero yields $c^* = \mathrm{Cov}(Y,X)/\mathrm{Var}(X)$. Substituting back, $\mathrm{Var}(\tilde{Y}(c^*)) = \mathrm{Var}(Y) - \mathrm{Cov}(Y,X)^2/\mathrm{Var}(X) = \mathrm{Var}(Y)(1-\rho^2)$ since $\rho^2 = \mathrm{Cov}(Y,X)^2/(\mathrm{Var}(Y)\,\mathrm{Var}(X))$. □

The variance reduction factor $1/(1-\rho^2)$ depends only on the correlation ρ between Y and the control X. A control with $|\rho| = 0.9$ reduces variance by a factor $1/(1-0.81) \approx 5.3$; with $|\rho| = 0.99$, by ≈ 50. Finding a good control is a creative problem: it must (a) have known expectation in closed form and (b) be highly correlated with Y.

4.2 Connection with OLS regression

The optimal coefficient c^* has a direct interpretation in terms of ordinary least squares. Consider regressing Y on X with an intercept:

$$Y \;=\; \alpha + \beta X + \varepsilon, \tag{11}$$

with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Cov}(\varepsilon, X) = 0$. The OLS slope estimator (in the population) is exactly

$$\beta^{\mathrm{OLS}} \;=\; \frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)} \;=\; c^*. \tag{12}$$

The control variate adjustment $\tilde{Y}(c^*) = Y - c^*(X - \mathbb{E}[X])$ is the *residual* of the OLS regression, shifted by the intercept $\alpha = \mathbb{E}[Y] - c^*\mathbb{E}[X]$ to recover the original mean. The variance of the residual is exactly $\mathrm{Var}(Y)(1-R^2)$ with $R^2 = \rho^2$, recovering (10).

This is more than a calculation curiosity: it tells us that *control variate variance reduction is just OLS regression applied to Monte Carlo.* Multiple control variates correspond to multiple regression, and the same intuition (more controls help if they bring genuine new information beyond the existing controls, i.e. they are not collinear with them) applies. We do not pursue multiple controls in this block, but the framework extends mechanically.

4.3 The empirical c^* and its bias

In practice $\mathrm{Cov}(Y,X)$ and $\mathrm{Var}(X)$ are unknown. We estimate them from the same M paths used for the Monte Carlo:

$$\hat{c} \;\stackrel{\text{def}}{=}\; \frac{\hat{\mathrm{Cov}}(Y,X)}{\hat{\mathrm{Var}}(X)} \;=\; \frac{\sum_{i=1}^{M}(Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^{M}(X_i - \bar{X})^2}, \tag{13}$$

which is exactly the OLS slope estimate from the sample. The estimator becomes

$$\hat{I}_M^{\mathrm{CV}} \;\stackrel{\text{def}}{=}\; \frac{1}{M}\sum_{i=1}^{M}\big(Y_i - \hat{c}\,(X_i - \mathbb{E}[X])\big). \tag{14}$$

This estimator is no longer strictly unbiased for I. Because $\hat{c}$ is itself a function of $\{(Y_i, X_i)\}_{i=1}^{M}$, its correlation with the $Y_i - \hat{c}\,(X_i - \mathbb{E}[X])$ terms induces a small bias.

Proposition 4.2 (Bias of the empirical CV estimator)**.** *Under standard moment conditions, the bias of* $\hat{I}_M^{\mathrm{CV}}$ *is* $O(1/M)$*, while its variance is* $\mathrm{Var}(Y)(1-\rho^2)/M + O(1/M^2)$*.*

The proof is a perturbation analysis: $\hat{c} = c^* + O_p(M^{-1/2})$, and the bias contribution at this order vanishes by orthogonality. See Glasserman [3, Section 4.1.3] for details.

The practical consequence: the bias decays at the same rate as the square root of the standard error decays, so for any M at which the standard error is acceptably small, the bias is also negligibly small (in fact, asymptotically negligible compared to the standard error itself, since bias is $O(1/M)$ while standard error is $O(1/\sqrt{M})$). For $M \geq 10^3$ in our typical setting, the bias is below 10^{-4}, well within rounding tolerance for option pricing.

The alternative is a *pilot run* of M_0 paths to estimate c^* followed by a production run of M paths with $\hat{c}$ frozen, which gives an exactly unbiased estimator. We do not adopt this in our implementation: the empirical $\hat{c}$ approach is simpler, the bias is negligible at our sample sizes, and the OLS interpretation is more transparent.

4.4 The two controls we use

For the European call under GBM, two natural controls present themselves:

Control 1 – the underlying: $X_1 \stackrel{\text{def}}{=} e^{-rT} S_T$, the discounted terminal price. Its expectation is $\mathbb{E}[X_1] = e^{-rT}\mathbb{E}[S_T] = e^{-rT} \cdot S_0 e^{rT} = S_0$, by the risk-neutral martingale property. This control is correlated with the call payoff $(S_T - K)^+$ but only modestly: the correlation breaks down where the payoff is zero (in the OTM region), so the linear relationship is weak. We expect $\rho \approx 0.5$–0.7 at the ATM point, giving a variance reduction factor of ≈ 1.4–2.0.

Control 2 – the asset-or-nothing: $X_2 \stackrel{\text{def}}{=} e^{-rT}\, S_T \mathbf{1}_{\{S_T > K\}}$, the discounted asset-or-nothing payoff. Its expectation has a closed Black-Scholes form

$$\mathbb{E}[X_2] \;=\; S_0\, \Phi(d_1), \tag{15}$$

with $d_1 = (\log(S_0/K) + (r + \sigma^2/2)T)/(\sigma\sqrt{T})$ as in the standard Black-Scholes formula. (This is the value of the asset-or-nothing call, which is a building block of the Black-Scholes derivation.) Its key property: *it activates on exactly the same paths as the call*, so its correlation with $(S_T - K)^+$ is much higher — typically $|\rho| > 0.95$. We expect a variance reduction factor of ≥ 10.

The disparity between the two controls illustrates a general principle: *a good control variate matches the "shape" of the target.* The call payoff is zero on $\{S_T < K\}$ and linear in S_T on $\{S_T \geq K\}$. The discounted asset matches the linear part but is nonzero everywhere. The asset-or-nothing matches both the zero region and the linear region.

5 When to use which technique

The two techniques apply in different circumstances and are not mutually exclusive. A pragmatic decision tree:

- *Is the payoff a monotone function of the underlying randomness?* If yes, antithetic variates are essentially free insurance: they cost one extra payoff evaluation per pair (so per-VxT cost factor 2) but always reduce variance for monotone payoffs.
- *Is there a tractable control variate with high correlation?* If yes, control variates can deliver order-of-magnitude reductions, far beyond what AV achieves. The challenge is finding the control: it requires knowing the closed-form expectation of something correlated with the target.

- *Both?* The two can be combined: first apply antithetic variates to construct Y^{AV}, then use a control variate on Y^{AV}. The combination is implemented in production quant systems but adds complexity; we do not pursue it in this portfolio.

For the European call under GBM, which is our running example, the asset-or-nothing control is so effective that AV is a marginal addition on top. We implement and validate both separately to illustrate the principles, then leave the combination as an exercise.

6 What we do not cover

For completeness, three other variance reduction techniques exist and are well-developed in the literature:

Stratified sampling. Partition the support of an input random variable into strata, sample a fixed (proportional) number from each stratum, and combine the strata estimators. Reduces the "between-strata" component of variance but not the "within-strata" component. Particularly powerful in low dimensions and for inputs where good strata can be designed (such as sampling from a single normal: stratifying its quantile space delivers near-deterministic convergence). See Glasserman [3, Chapter 4].

Importance sampling. Change the probability measure to oversample from regions where the payoff is large, then reweight to recover an unbiased estimator. Mathematically a Radon–Nikodym change of measure. Most effective when the payoff has a localised "important" region (deep out-of-the-money options, rare events). See Glasserman [3, Chapter 4], Asmussen & Glynn [1].

Multilevel Monte Carlo (MLMC). Combine simulations at multiple discretisation grids, exploiting that fine-grid simulations are expensive but coarse-grid simulations are noisy. Reduces the total cost to achieve a target accuracy for SDE-based MC. See Giles [2].

We do not cover these in the portfolio for reasons of scope: the asset-or-nothing control variate alone delivers more variance reduction than this entire course needs, and the techniques above become valuable in regimes (deep OTM, very long-dated, exotic path-dependent) that we do not target here.

7 Bibliographic notes

The textbook reference for variance reduction in financial Monte Carlo is Glasserman [3], Chapter 4. The OLS connection of control variates is standard, well-treated in any book on simulation theory, e.g. Asmussen & Glynn [1], Chapter V. The asset-or-nothing as a Black-Scholes building block is in Hull [4], Chapter 26. The general framework of variance-times-time as the right comparison criterion is folklore in the simulation community and stated explicitly in Glasserman [3, Section 1.1.3].

References

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